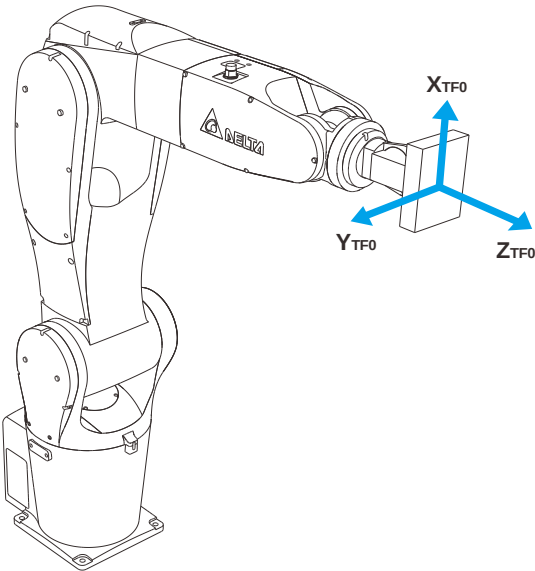


4.7.4 How to teach the Tool frame directions

Set the coordinate frame (Mode) to World (both the User Frame ID and Tool Frame ID are set to 0 in this mode), and then follow these steps to convert the default Tool Frame Z axis direction for it to align with the direction of the end effector.



Default Tool Frame directions

- 1. Click **Determine** in the Tool Orientation area to open the Tool Orientation Calculation screen.

Tool Orientation

Input :

(1) Enter the tool orientation data (Unit : Degree)

| Roll (RX) | Pitch (RY) | Yaw (RZ) |
|-----------|------------|----------|
| 0.000     | 0.000      | 0.000    |

(2) Determine the tool orientation data by teaching

Determine

Tool Orientation Set

- 2. Select a fixed point, and then move the end effector to the fixed point. Select the **Origin** table row, and then click **Teach** to save the current position to the row.

Origin

Tool Orientation Calculation

ID : 1

Input :

Enter data and Calculate

|          | X (mm)  | Y (mm) | Z (mm)  |
|----------|---------|--------|---------|
| Origin   | 444.001 | 0.000  | 211.321 |
| + X axis | 0       | 0      | 0       |
| + Y axis | 0       | 0      | 0       |

Teach

Tool Orientation Set

Get

Output :

| Roll (RX) | Pitch (RY) | Yaw (RZ) |
|-----------|------------|----------|
| 0         | 0          | 0        |